

Projeto 4 - Tempo de resposta

Etapa 1: Derivada em $r=4$

$$\begin{aligned}f(r) &= \sqrt{r} & f''(r) &= -\frac{1}{4r^{3/2}} & f''(4) &= -\frac{1}{4 \cdot 8} = -\frac{1}{32} & f'''(r) &= \frac{3}{8r^{5/2}} & f'''(4) &= \frac{3}{8 \cdot 64} = \frac{3}{512} \\f(4) &= \sqrt{4} = 2 & f''(4) &= -\frac{1}{4(4^{3/2})} = -\frac{1}{4(8)} = -\frac{1}{32} & f'''(4) &= \frac{3}{8 \cdot 64} = \frac{3}{512} \\f'(r) &= \frac{1}{2\sqrt{r}} & f''(4) &= -\frac{1}{32} & f'''(4) &= \frac{3}{512} \\f'(4) &= \frac{1}{4} \\f''(4) &= -\frac{1}{32}\end{aligned}$$

$$\text{Final: } f(4) = 2, f'(4) = \frac{1}{4}, f''(4) = -\frac{1}{32}, f'''(4) = \frac{3}{512}$$

$$\frac{3}{512}$$

Etapa 2: Montagem de $P_3(x)$

$$P_3(r) = f(4) + f'(4)(r-4) + \frac{f''(4)}{2!}(r-4)^2 + \frac{f'''(4)}{3!}(r-4)^3$$

$$P_3(r) = 2 + \frac{1}{4}(r-4) - \frac{1}{32 \cdot 2}(r-4)^2 + \frac{3}{512 \cdot 6}(r-4)^3$$

$$P_3(r) = 2 + \frac{1}{4}(r-4) - \frac{1}{64}(r-4)^2 + \frac{1}{512}(r-4)^3$$

Etapa 3: Estimativa em $r=4,2$

$$P_3(r) = 2 + \frac{1}{4}(r-4) - \frac{1}{64}(r-4)^2 + \frac{1}{512}(r-4)^3$$

$$P_3(4,2) = 2 + \frac{1}{4}(4,2-4) - \frac{1}{64}(4,2-4)^2 + \frac{1}{512}(4,2-4)^3$$

$$4,2-4=0,2$$

$$P_3(4,2) = 2 + \frac{1}{4}(0,2) - \frac{1}{64}(0,2)^2 + \frac{1}{512}(0,2)^3$$

$$= 2 + 0,05 - 0,000625 + 0,00015625$$

$$P_3(4,2) = 2,04939$$

Etapa 4: Comparação com o valor real e erro

$$P_3(4,2) = 2,04939$$

$$I(4,2) = \sqrt{4,2}$$

$$I(4,2) = 2,049390153$$

$$\text{Erro} = \text{Valor real} - \text{Estimativa}$$

$$\text{Erro} = 2,049390153 - 2,04939$$

$$\text{Erro} = 0,000000153$$

Etapa 5: Interpretação no contexto do Traca Ticket

$$T(r) = \sqrt{r}$$

$$T(4,2) = 2,04939$$

$$0,000000153$$